

Matrix Algebra

Geometric Algorithms Lecture 9

Practice Problem

Determine the matrix that implements the transformation which projects vectors in $\mathbb{R}^2$ vertically onto the line $y = 2x$ in $\mathbb{R}^2$

(Show that the transformation which projects onto $y = 2x + 3$ is not linear)

Answer

Objectives

- » Connect questions about **matrix equations** and **linear transformations**
- » Motivate **matrix multiplication**
- » Define matrix multiplication
- » Look at the **algebra** of matrix multiplication

Keywords

one-to-one transformation

onto transformation

matrix multiplication

row-column rule

matrix addition and scaling

non-commutativity

Recap

Recap: Matrix Transformations

Matrices *transform* vectors into vectors the span of its columns

$$\mathbf{x} \mapsto A\mathbf{x}$$

map a vector $\mathbf{v}$ to the vector $A\mathbf{v}$

Recap: Linear Transformations

Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Matrix transformations are linear transformations

Recap: Matrix of Linear Transformation

Thm. A transformation T is linear if and only if it is a matrix transformation (there is a matrix that "implements" T)

Linear transformations are exactly matrix transformations

Recap: Matrix of Linear Transformation

Thm. For any linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$,
the matrix

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \dots \quad T(\mathbf{e}_n)]$$

is the unique matrix that implements T

How do matrix properties relate
to transformation properties?

One-to-One and Onto

Recap: New Interpretations

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A
transforms into $\mathbf{b}$?

Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A
transforms into $\mathbf{b}$

More New Interpretations

Does $A\mathbf{x} = \mathbf{b}$ have a solution for any choice of $\mathbf{b}$?

Does $A\mathbf{x} = \mathbf{0}$ have a unique solution?

More New Interpretations

Does $A\mathbf{x} = \mathbf{b}$ have at least one solution for any choice of $\mathbf{b}$?

Does $A\mathbf{x} = \mathbf{b}$ have at most one solution for any choice of $\mathbf{b}$?

Why?

$A\mathbf{x} = \mathbf{0}$ has a
unique solution

$\equiv$

$A\mathbf{x} = \mathbf{b}$ has at most one
solution

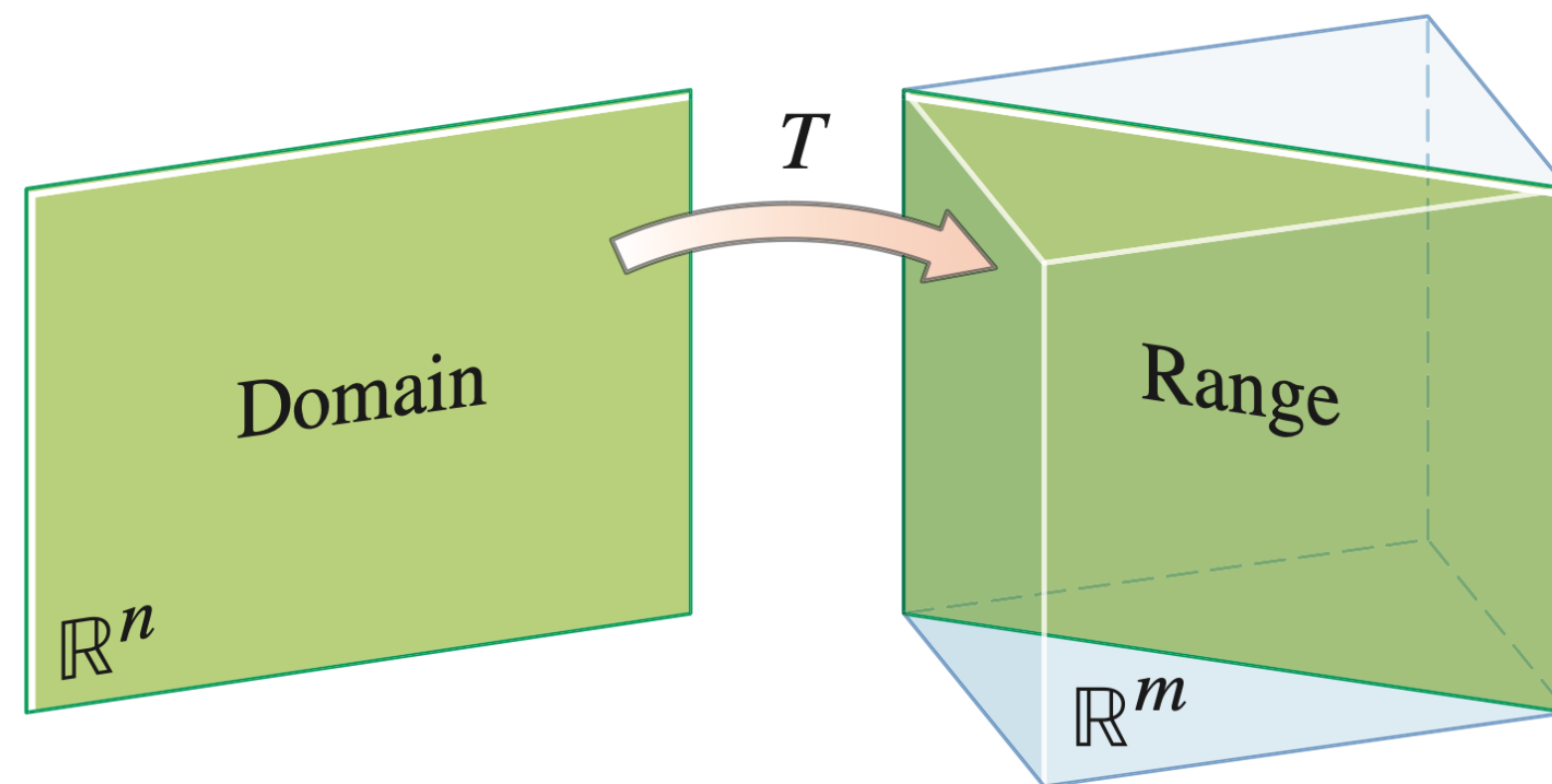
Onto Transformations

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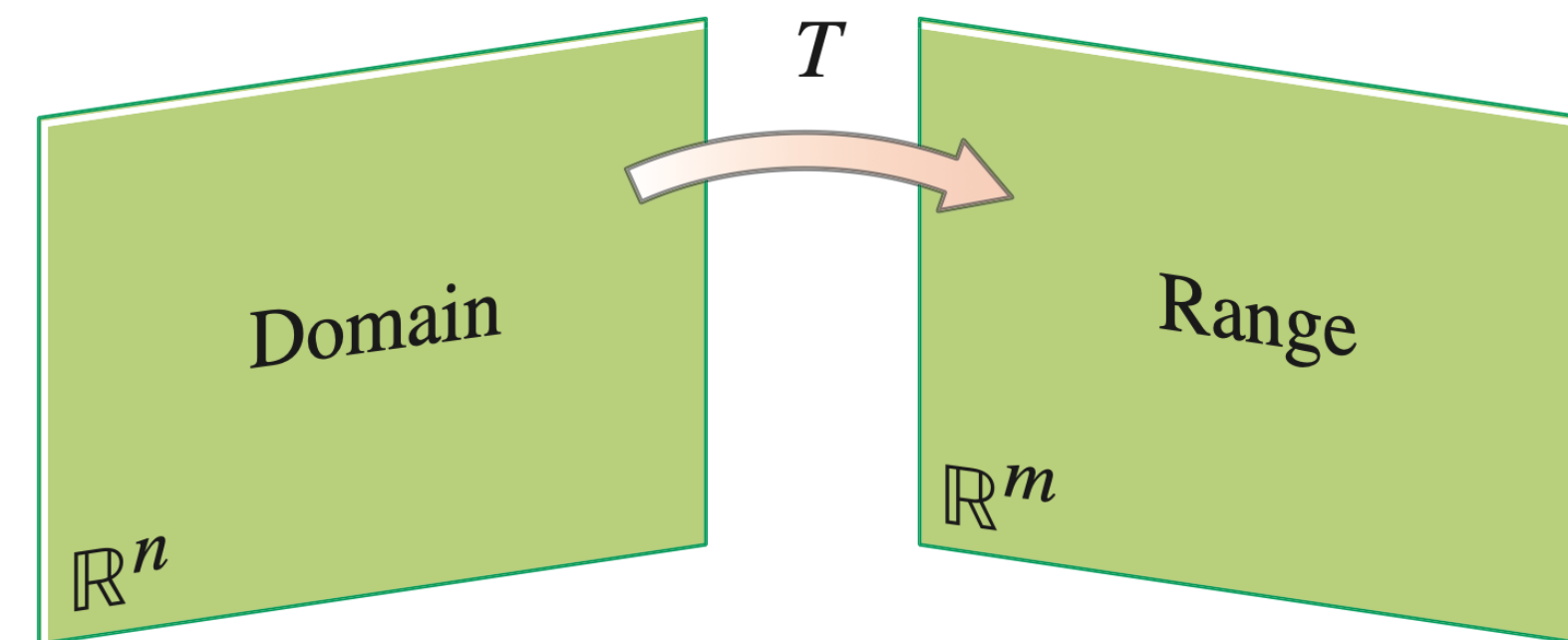
Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is ***onto*** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the **image of at least one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

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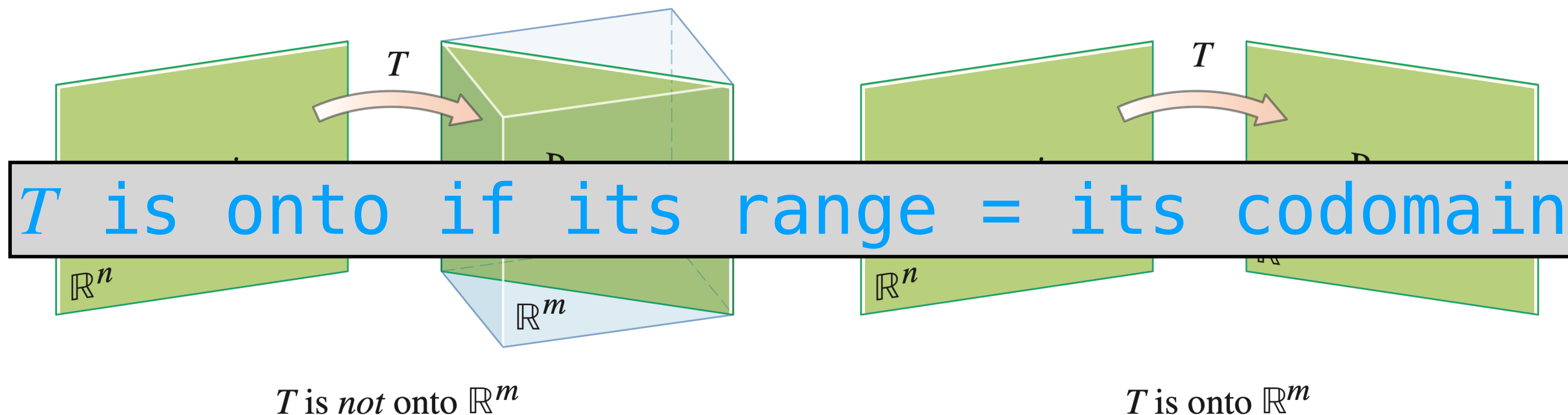
T is *not* onto $\mathbb{R}^m$



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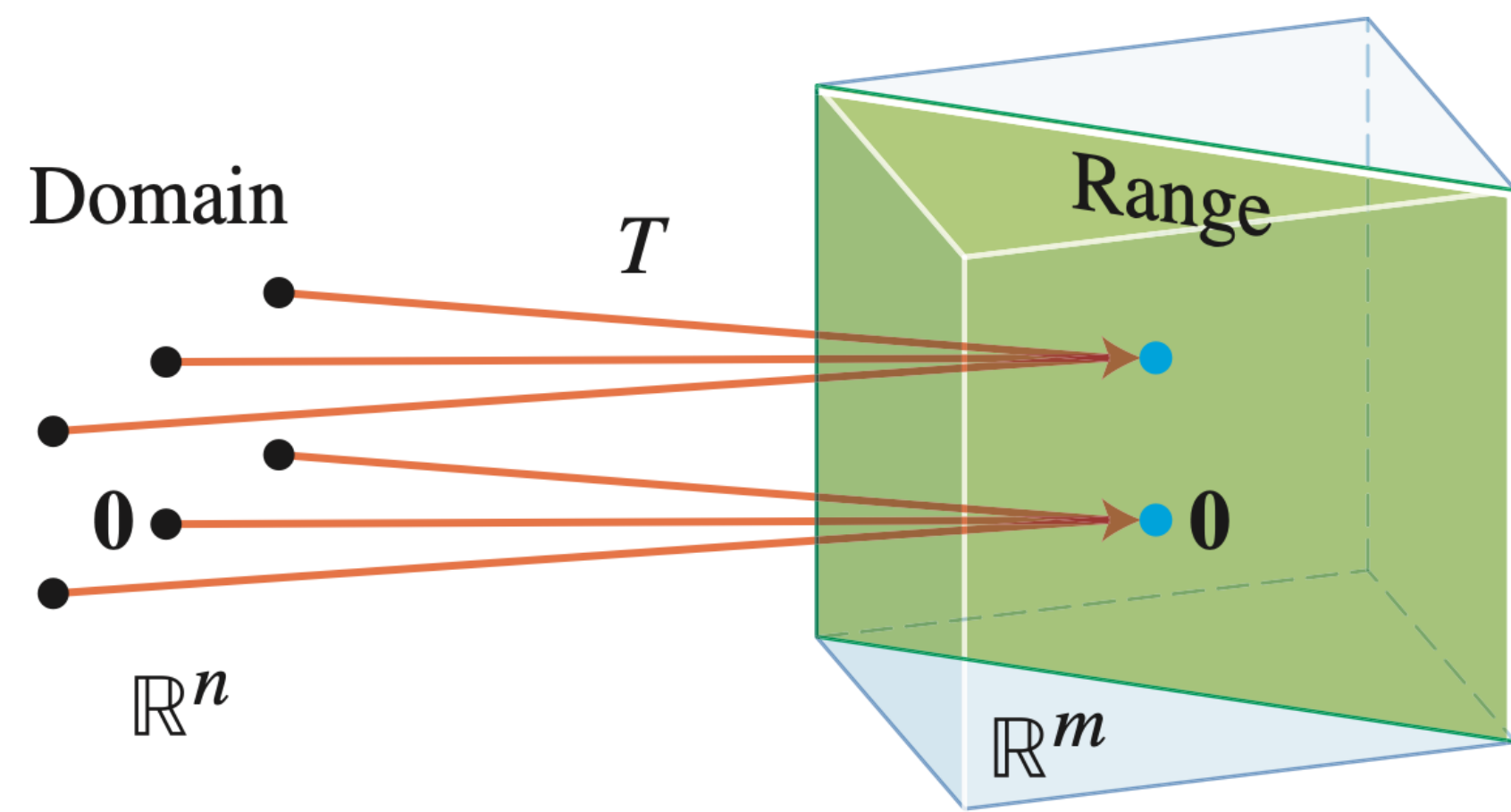
One-to-one Transformations

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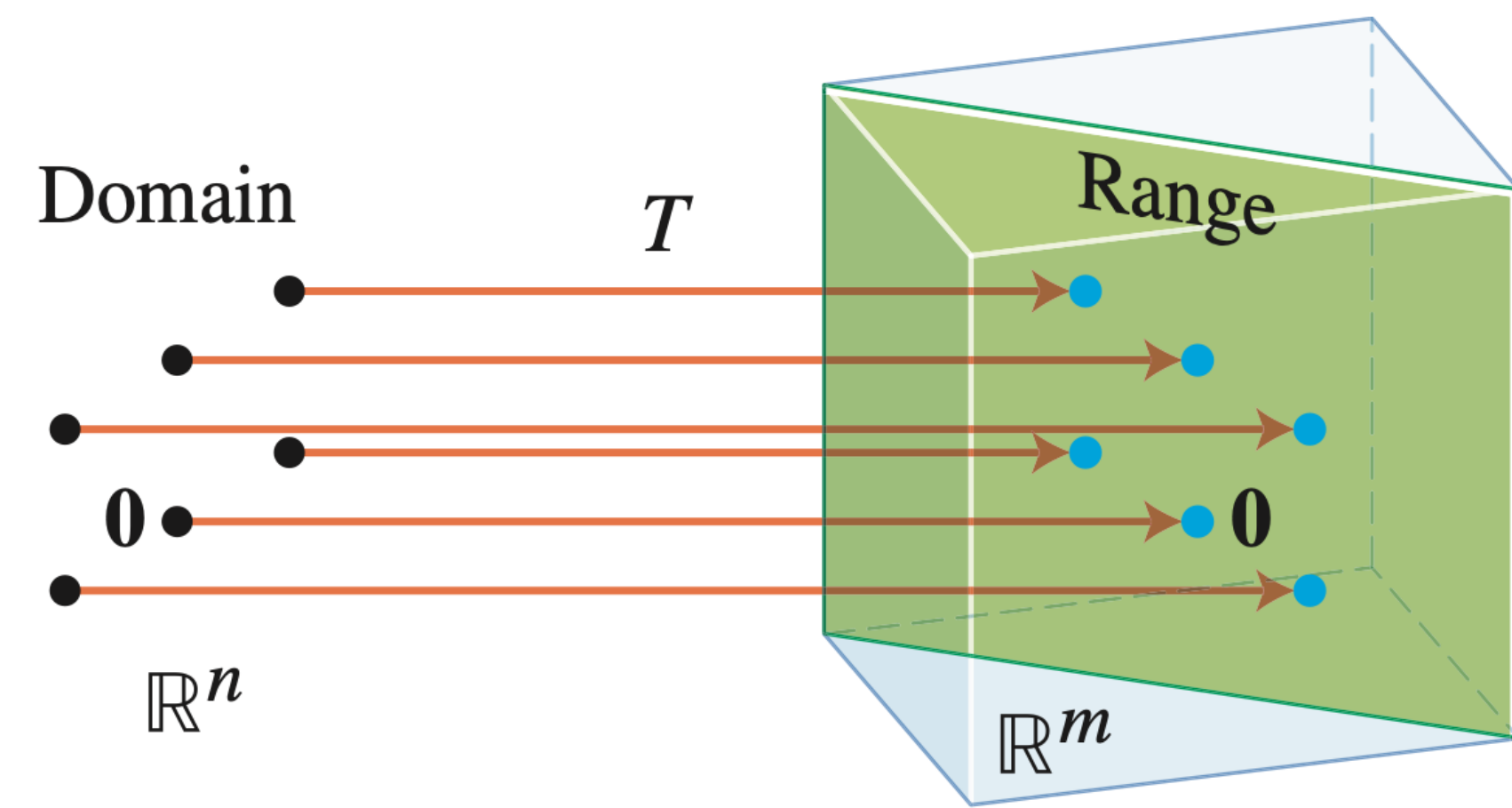
Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is ***one-to-one*** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the **image of at most one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

One-to-one Transformations

Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the **image of at most one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

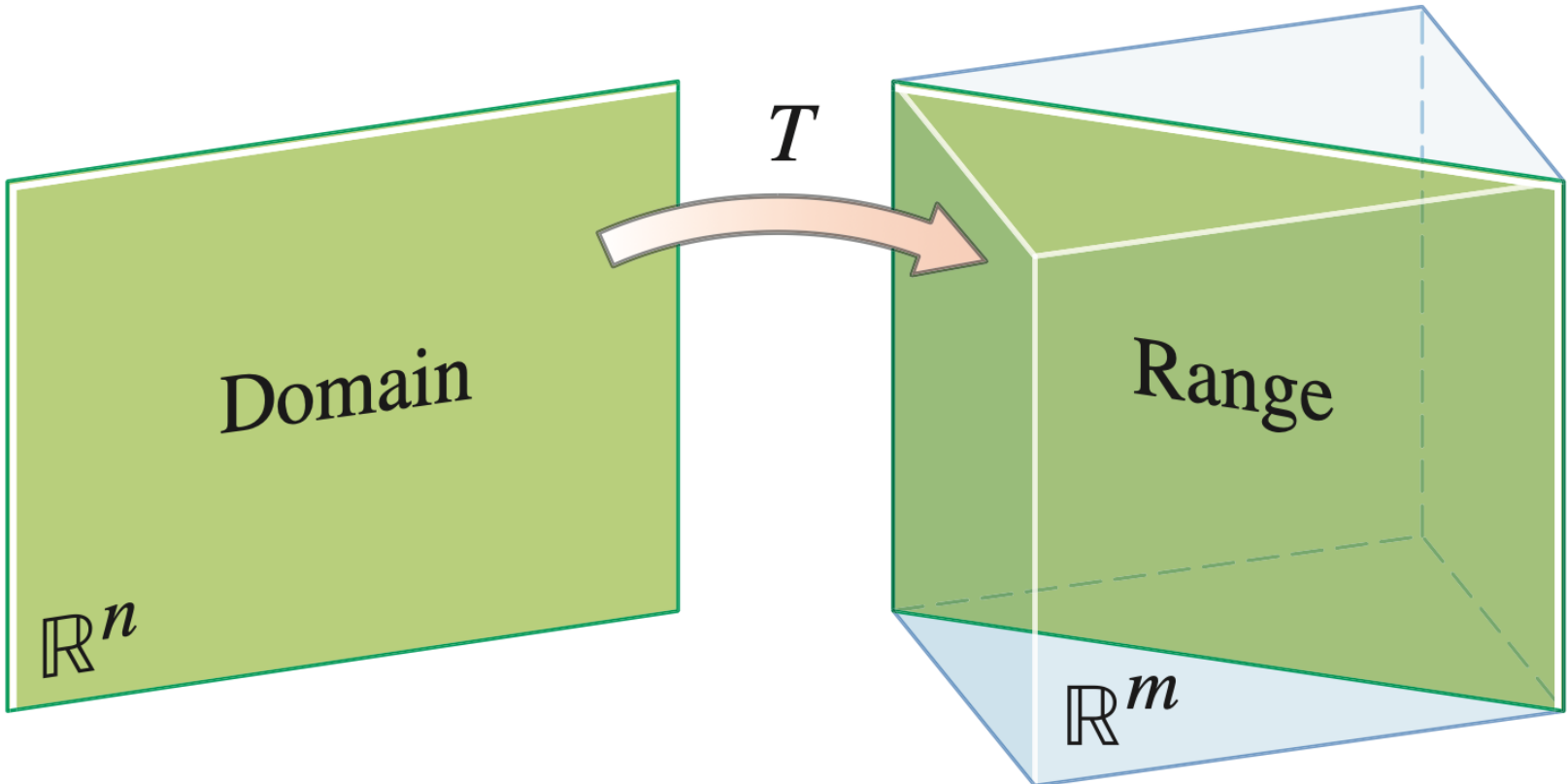


T is *not* one-to-one

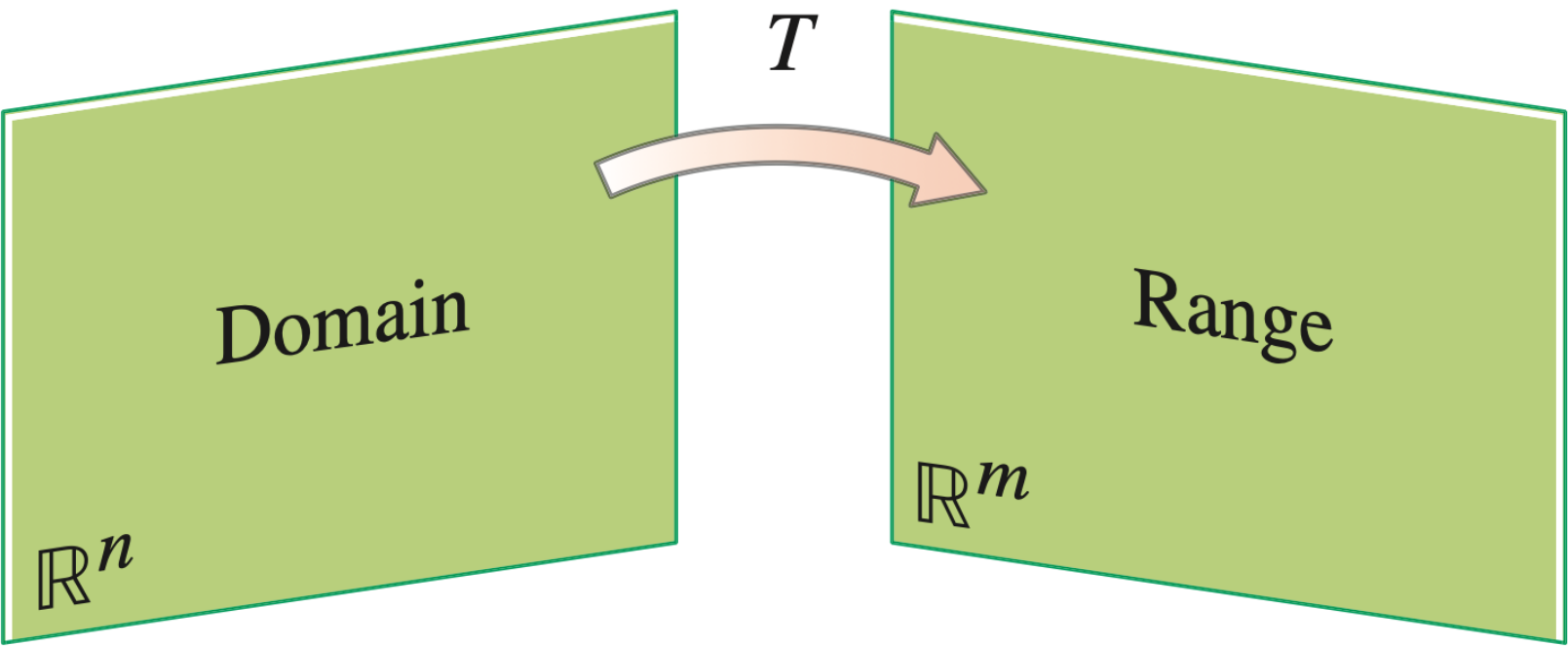


T is one-to-one

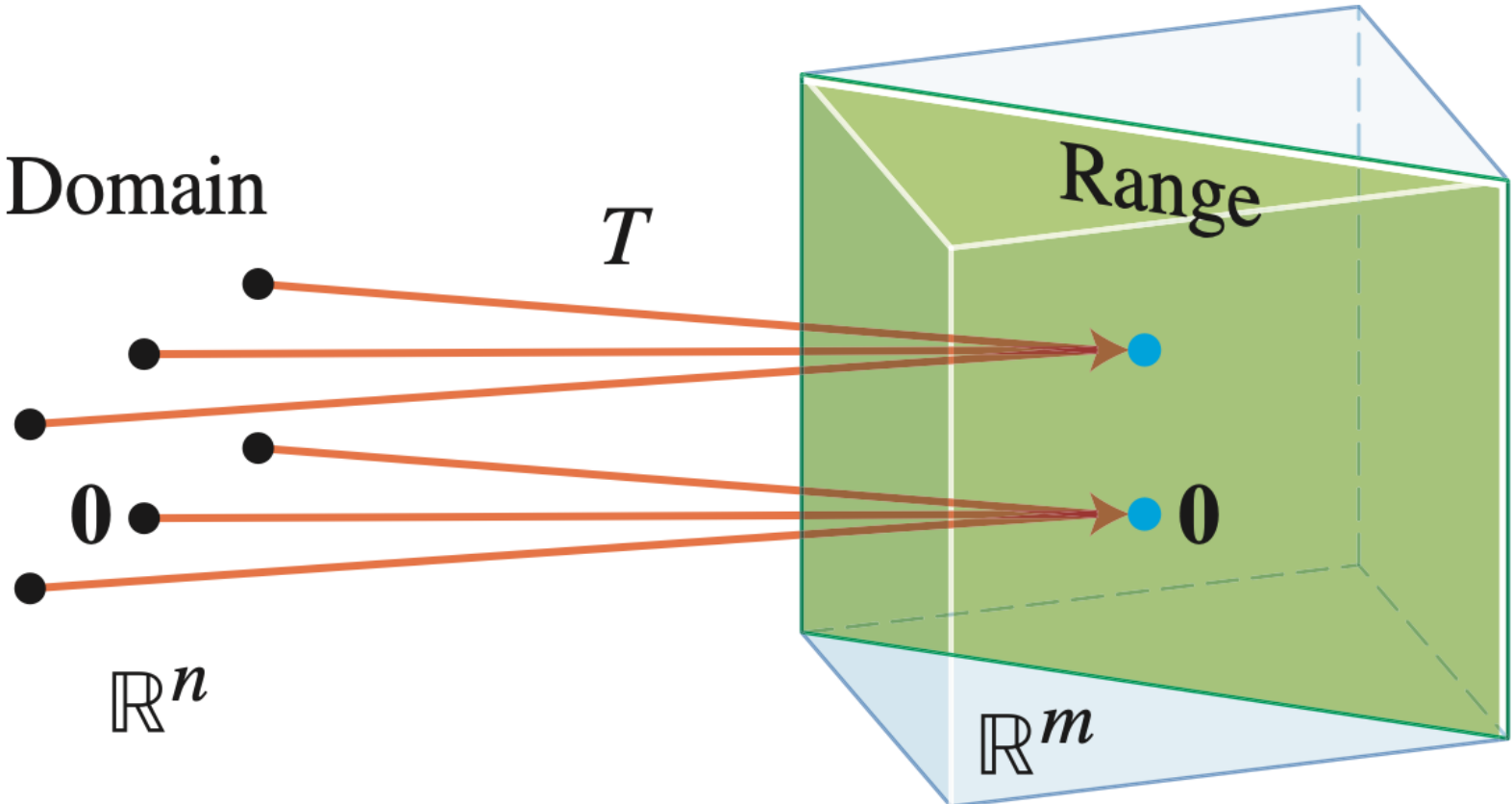
Comparing Pictures



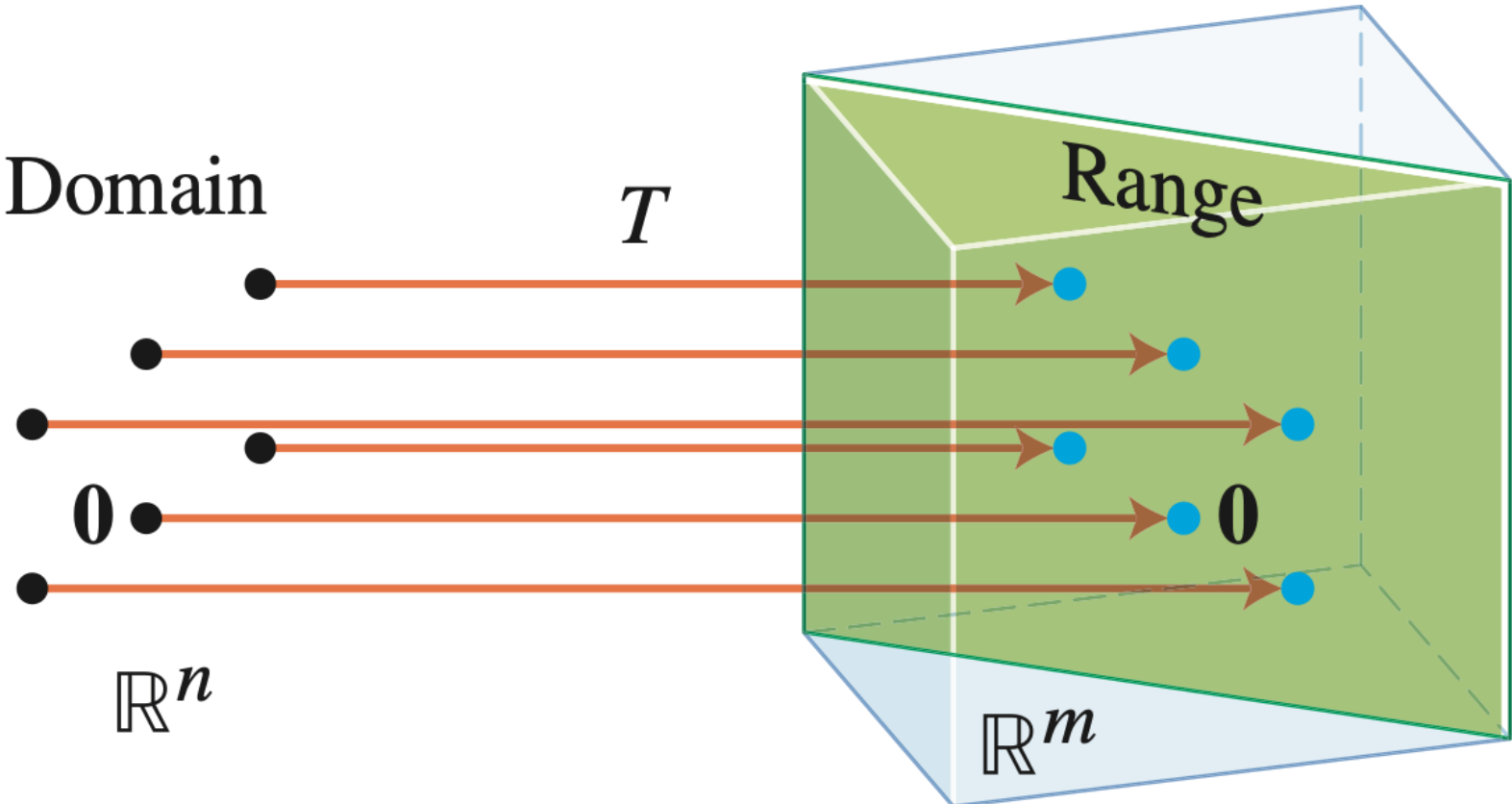
T is not onto $\mathbb{R}^m$



T is onto $\mathbb{R}^m$



T is not one-to-one



T is one-to-one

Example: 1-1 and Onto

$$\mathbf{x} \mapsto \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \mathbf{x}$$

Example: 1-1 and not Onto

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}$$

Example: Not 1-1 and not Onto

$$\mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{x}$$

Example: Not 1-1 and Onto

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Taking Stock: Onto

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 $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

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» $\text{range}(T) = \text{codomain}(T)$

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- » the columns of A span $\mathbb{R}^m$

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- » the columns of A span $\mathbb{R}^m$
- » A has a pivot position in every row

Taking Stock: One-to-One

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- » T is one-to-one
- » $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
- » $A\mathbf{x} = \mathbf{0}$ has only the trivial solution

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- » The columns of A are linearly independent
- » A has a pivot position in every column

How To: One-to-One and Onto

Question. Show that the linear transformation T is one-to-one/onto

Solution. (one approach) Find the matrix which implements T and see if it has a pivot in every column/row

Warning: this is not the only way. Always try to think if you can solve it using *any* of the perspectives

Example: 1-1 and Onto

$$\mathbf{x} \mapsto \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \mathbf{x}$$

Example: 1-1 and not Onto

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}$$

Example: Not 1-1 and not Onto

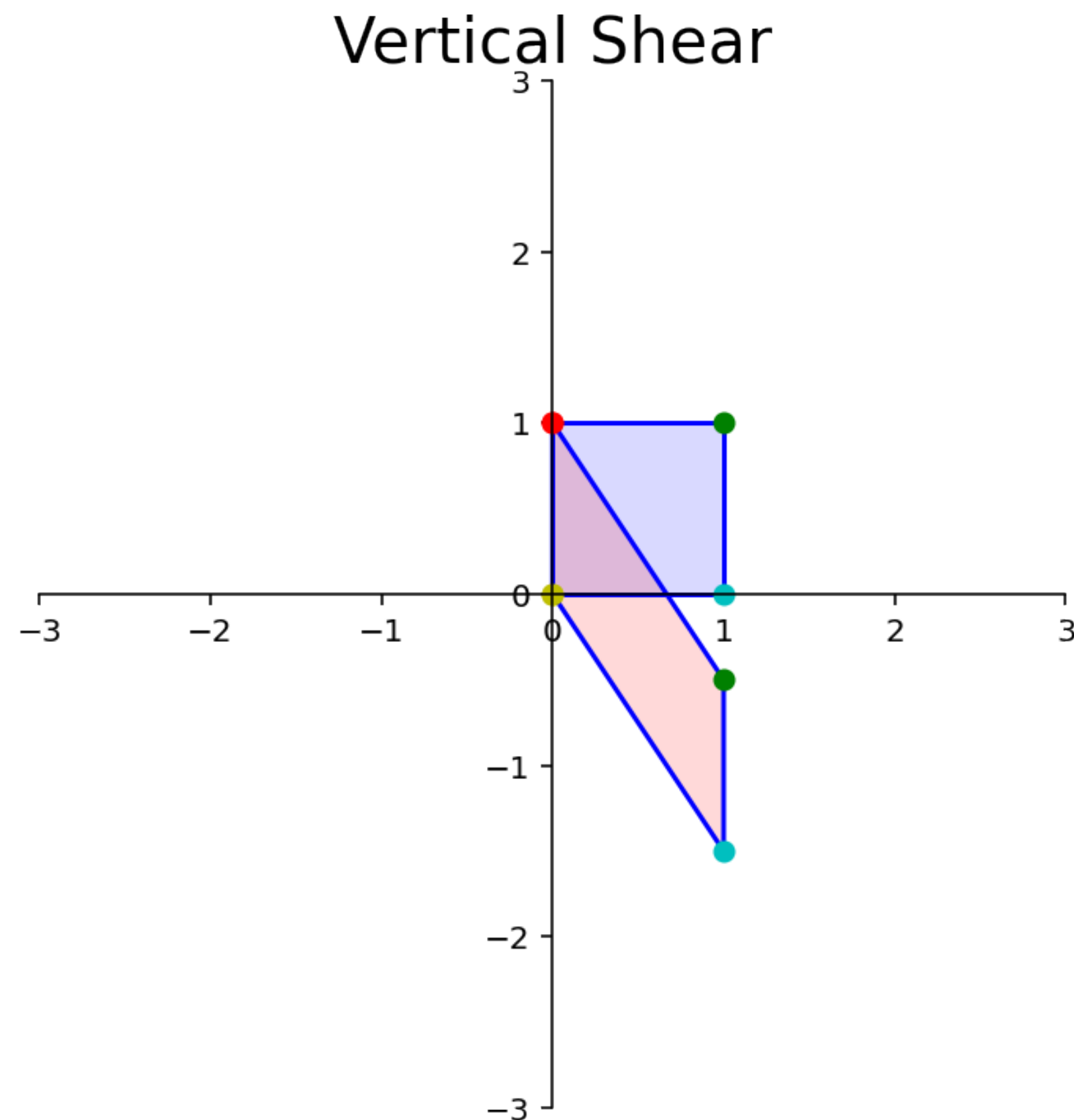
$$\mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{x}$$

Example: Not 1-1 and Onto

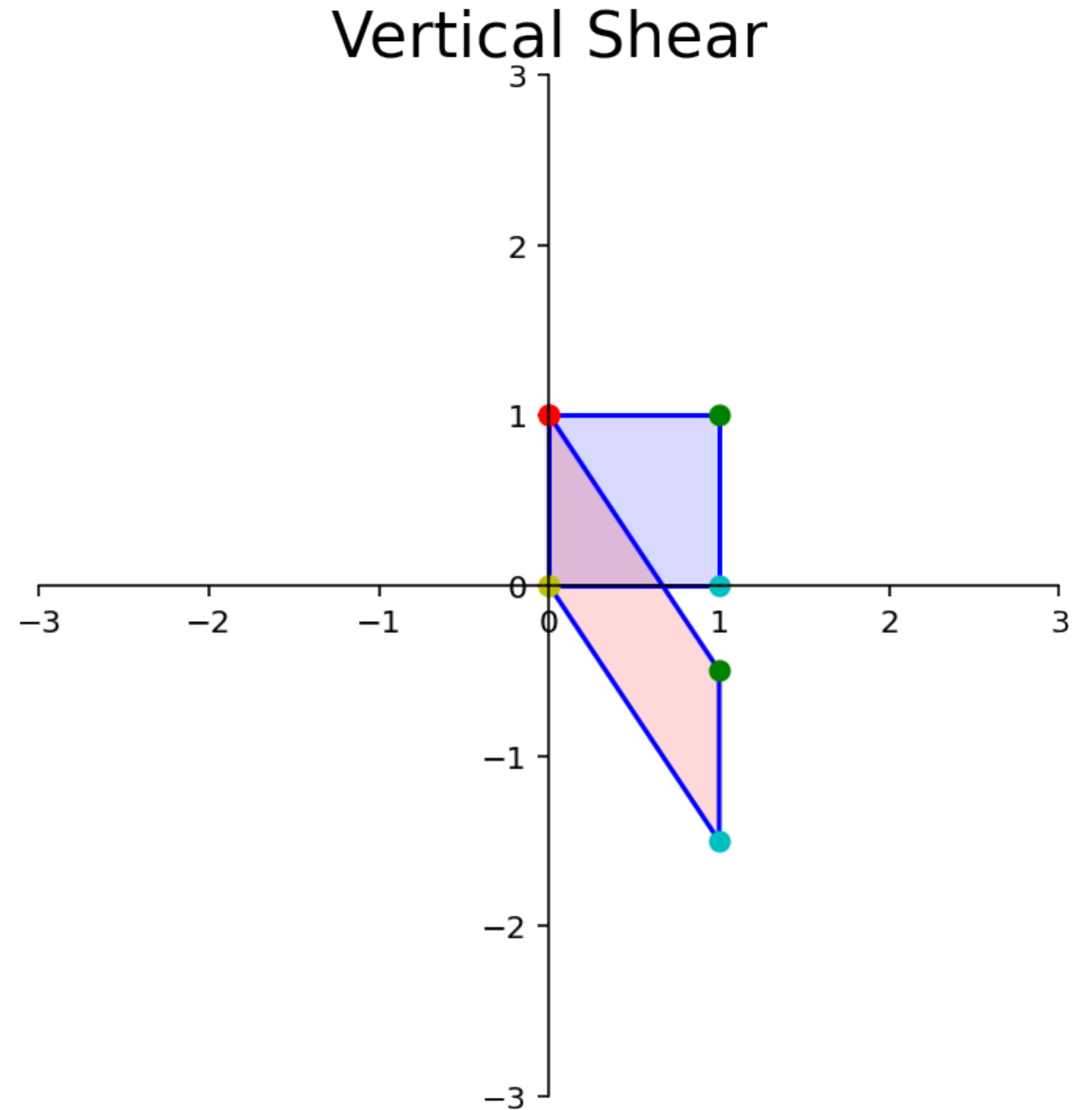
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Question

Is vertical shearing a 1-1 transformation?
Justify your answer

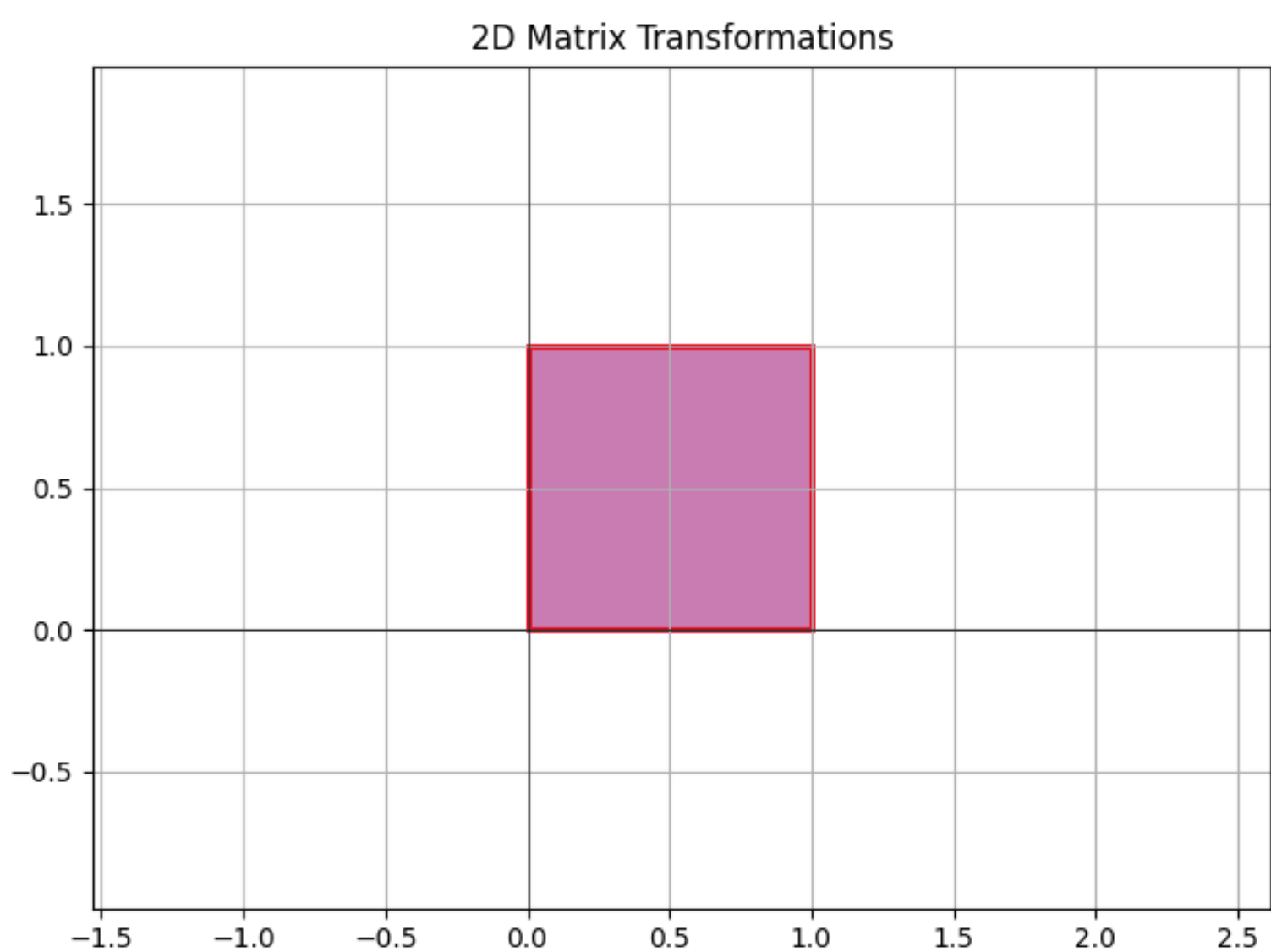


Answer: Yes

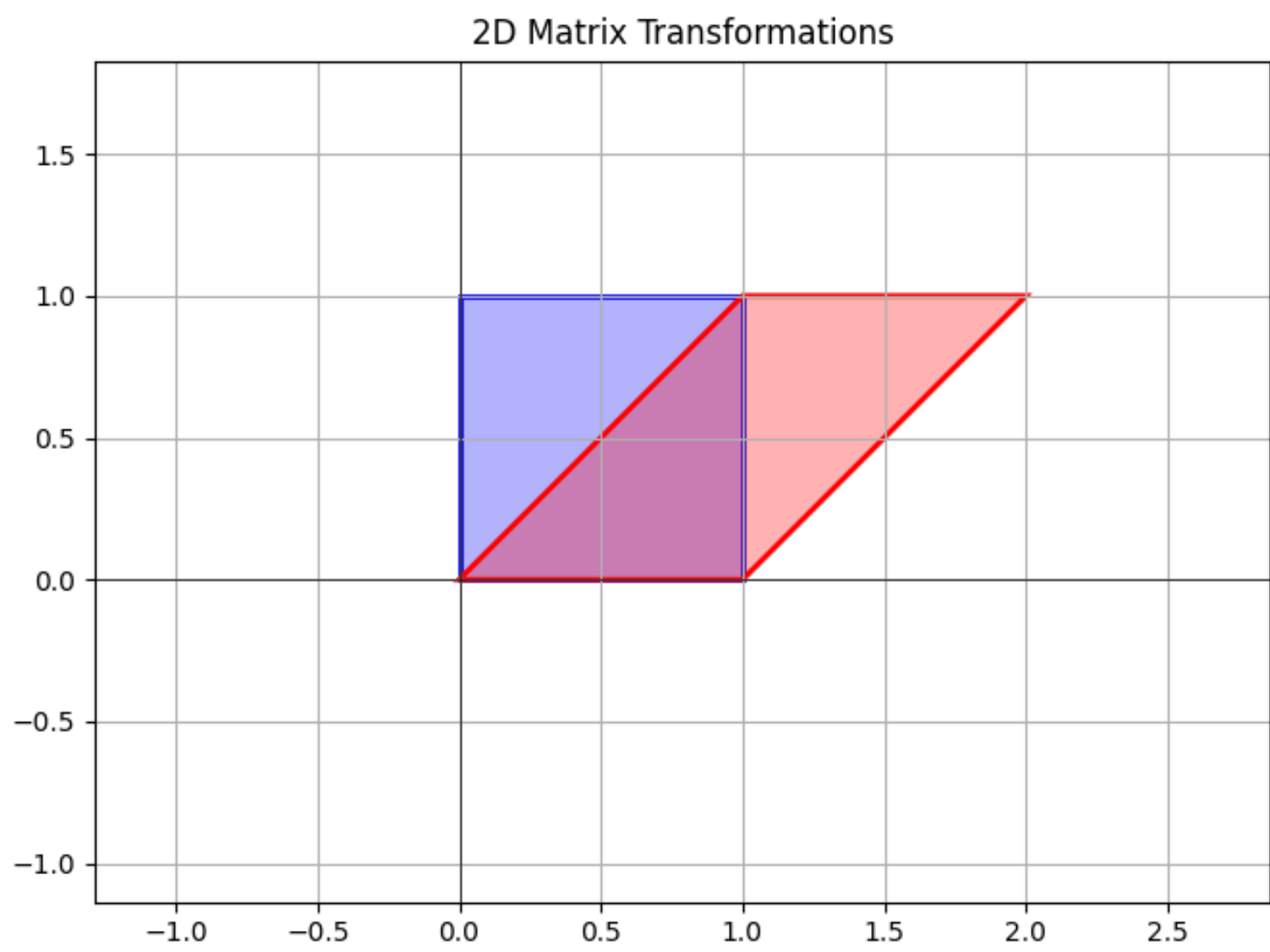


Composing Linear Transformations

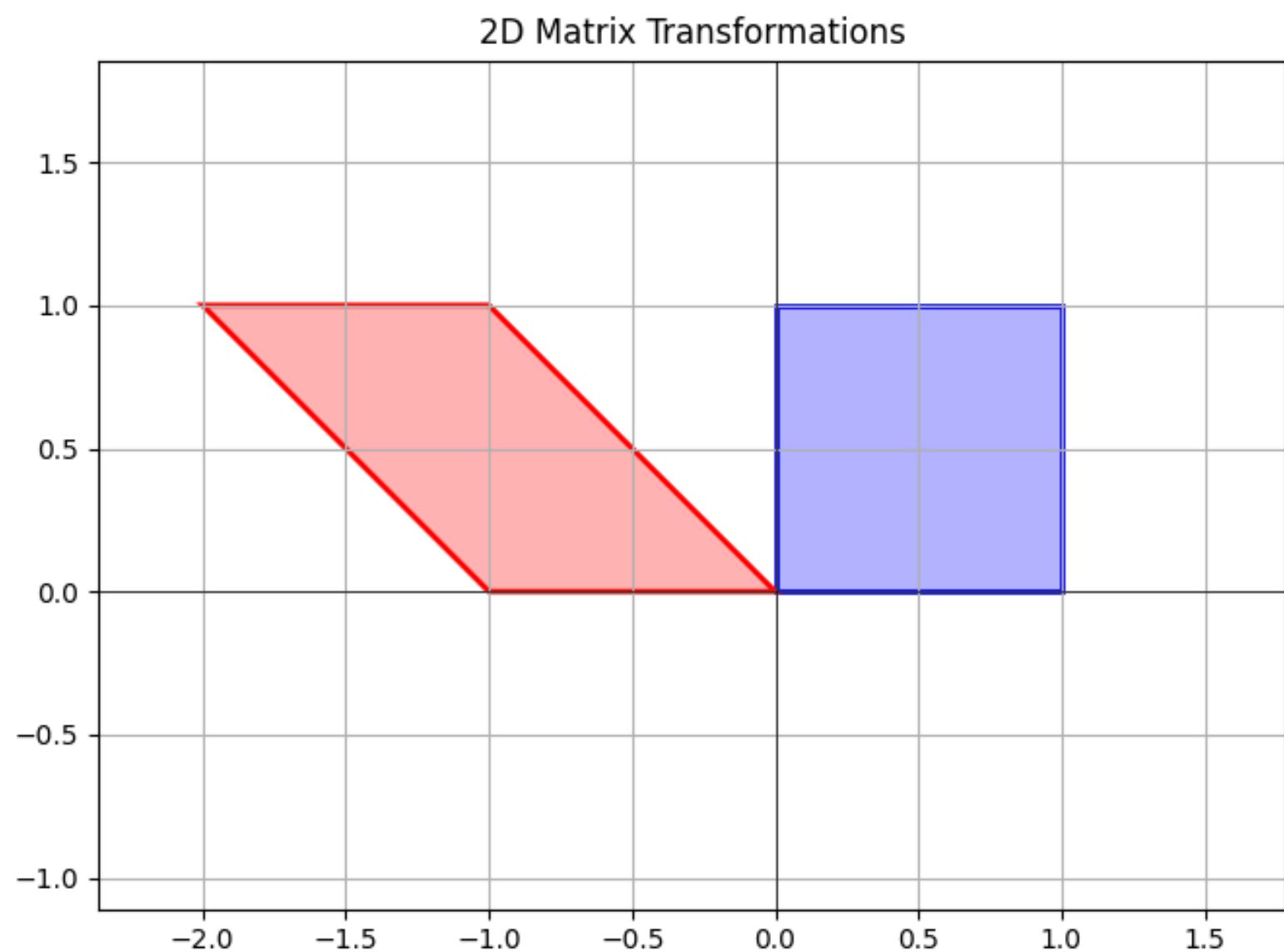
Shearing and Reflecting (Geometrically)



shear



reflect

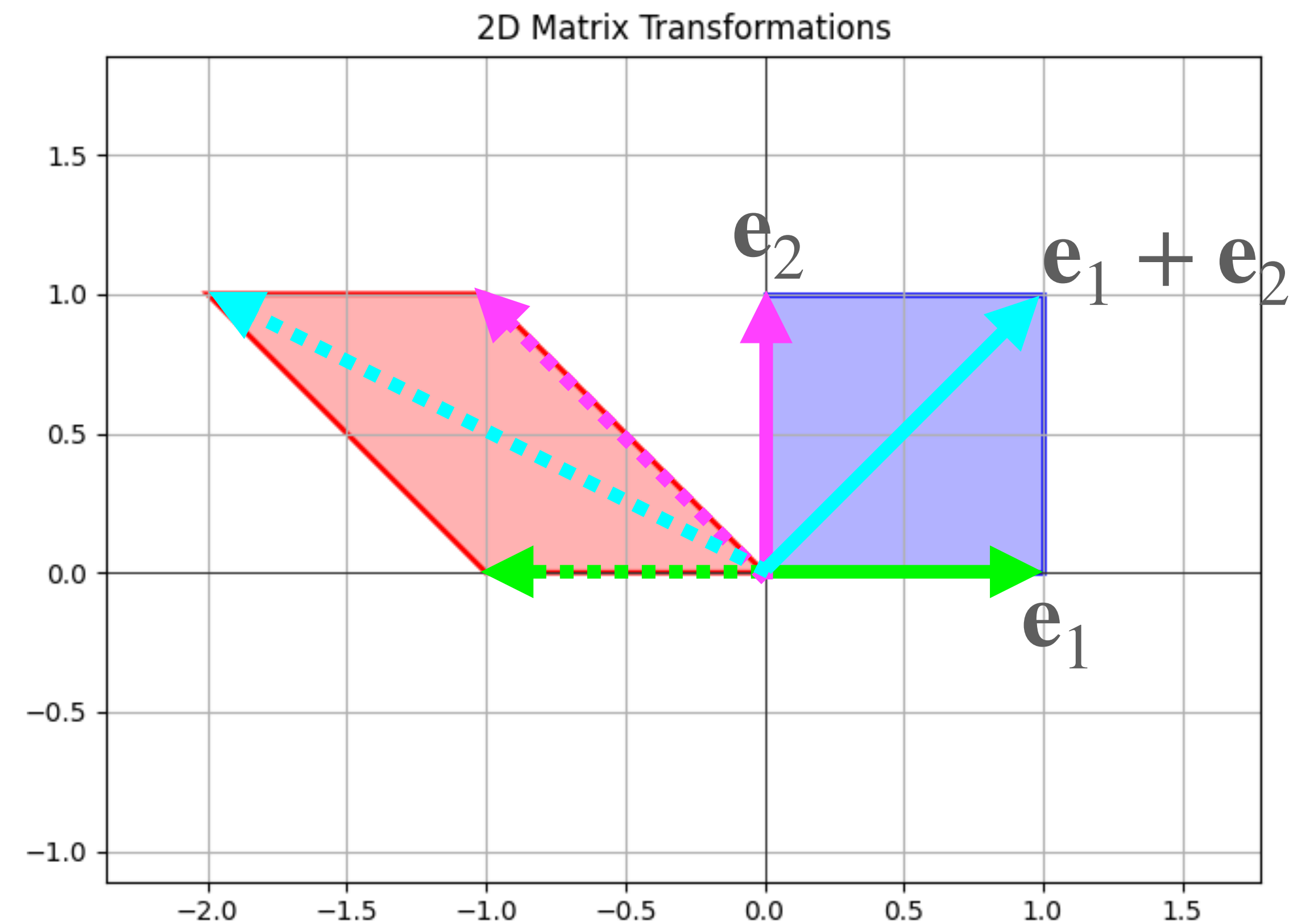


Shearing and Reflecting Matrix

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \mapsto$$



Shearing and Reflecting (Algebraically)

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right)$$

reflect shear

First multiply by shear matrix, then multiply
by reflection matrix

Shearing and Reflecting (Algebraically)

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right)$$

reflect shear

First multiply by shear matrix, then multiply
by reflection matrix

This gives us the same transformation

Shearing and Reflecting

$$\begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x} \right)$$

The Key Fact

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Thm. The composition of two linear transformations is a linear transformation

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This means the composition of two matrix transformations can be represented as a *single* matrix

The Key Question

*Given two linear transformations,
how to we compute the matrix which
implements their composition?*

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Matrix Multiplication

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Shearing and Reflecting

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) =$$

General Composition (2D)

$$A \left(\begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) =$$

Matrix Multiplication

Def. For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ with columns $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p$ the product $AB \in \mathbb{R}^{m \times p}$ is given by

$$AB = A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \dots & A\mathbf{b}_p \end{bmatrix}$$

Replace each column of B with A multiplied by that column

Tracking Dimensions

This only works if the number of columns of the left matrix matches the number of rows of the right matrix

The diagram illustrates matrix multiplication with dimension tracking. It shows three matrices arranged horizontally, separated by an equals sign. The first matrix is a 5x3 matrix, represented by a blue vertical line on the left labeled m and a red horizontal line on top labeled n . The second matrix is a 3x4 matrix, represented by a red vertical line on the left labeled n and a purple horizontal line on top labeled k . The third matrix is a 5x4 matrix, represented by a blue vertical line on the left labeled m and a purple horizontal line on top labeled k . Each matrix contains asterisks representing elements. Below each matrix, its dimensions are written in a colored box: $(m \times n)$ for the first, $(n \times k)$ for the second, and $(m \times k)$ for the third.

$$\begin{matrix} m \\ \left[\begin{array}{ccc} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{array} \right] \end{matrix} \begin{matrix} n \\ \left[\begin{array}{cccc} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{array} \right] \end{matrix} = \begin{matrix} m \\ \left[\begin{array}{cccc} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{array} \right] \end{matrix}$$

$(m \times n)$ $(n \times k)$ $(m \times k)$

Important

Even if AB is defined, it may
be that BA is *not* defined

Non-Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \left[\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right]$$

Non-Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \left[\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right]$$

These are not defined

Example

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} \end{bmatrix}$$

The Key Fact (Restated)

For any matrices A and B such that AB is defined and any vector $\mathbf{v}$

$$A(B\mathbf{v}) = (AB)\mathbf{v}$$

The matrix implementing the composition is the product of the two underlying matrices

Row-Column Rule

Given $A \in \mathbb{R}^{m \times n}$ and $n \times p$ matrix $B \in \mathbb{R}^{n \times p}$ the entry $(AB)_{ij}$ is given as:

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Example

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} =$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with its top row highlighted in light blue. The second matrix is a 3x4 matrix with its first column highlighted in light red. An equals sign follows, and then a 5x4 matrix where the top-left element is highlighted in light purple, representing the result of multiplying the first row of the first matrix by the first column of the second matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

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$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

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Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with its first row highlighted in light blue. The second matrix is a 3x4 matrix with its third column highlighted in light red. An equals sign follows, and then a 5x4 matrix where the element at the intersection of the first row and third column is highlighted in light purple, representing the result of the dot product of the first row of the first matrix and the third column of the second matrix.

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

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Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with its first row highlighted in light blue. The second matrix is a 3x4 matrix with its fourth column highlighted in light red. An equals sign follows, and then a 5x4 matrix where the element in the first row and fourth column is highlighted in light purple, representing the result of the dot product of the first row of the first matrix and the fourth column of the second matrix.

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements marked with asterisks (*). The second matrix is a 3x4 matrix, also with all elements marked with asterisks. The third matrix is a 5x4 matrix, also with all elements marked with asterisks. The second matrix is positioned between the first and third matrices, with an equals sign (=) to its right. The first matrix has its second row highlighted in light blue. The second matrix has its first column highlighted in light red. The third matrix has its first row highlighted in light purple. This visualizes the calculation of the element in the second row and first column of the product matrix AB, which is the dot product of the second row of A and the first column of B.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements represented by asterisks (*). The second matrix is a 3x4 matrix, also with all elements as asterisks, but its second column is highlighted with a light red background. An equals sign (=) follows. The third matrix is a 5x4 matrix with all elements as asterisks, but its second column is highlighted with a light purple background. This illustrates that the second row of the first matrix multiplied by the second column of the second matrix results in the second column of the third matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements represented by asterisks (*). The second matrix is a 3x4 matrix, also with all elements represented by asterisks. The third matrix is a 5x4 matrix, also with all elements represented by asterisks. An equals sign (=) is placed between the second and third matrices. The first matrix has its second row highlighted in light blue. The second matrix has its third column highlighted in light red. The third matrix has its third column highlighted in light purple, indicating the result of the dot product of the second row of the first matrix and the third column of the second matrix.

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$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; its third row is highlighted in light blue. The second matrix is a 3x4 matrix with asterisks in each cell; its first column is highlighted in light red. An equals sign follows. The third matrix is a 5x4 matrix with asterisks in each cell; its first row is highlighted in light purple. This visualizes the calculation of the element in the third row and first column of the product matrix.

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$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 grid of asterisks with its third row highlighted in light blue. The second matrix is a 3x4 grid of asterisks with its fourth column highlighted in light red. An equals sign follows, and then a 5x4 grid of asterisks where the element at the intersection of the third row and fourth column is highlighted in light purple.

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$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

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Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; the bottom row is highlighted in light blue. The second matrix is a 3x4 matrix with asterisks; the second column is highlighted in light red. An equals sign follows. The third matrix is a 5x4 matrix with asterisks; the element in the bottom row and second column is highlighted in light purple, representing the dot product of the bottom row of the first matrix and the second column of the second matrix.

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$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Example

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} =$$

Matrix Operations

Matrix-Vector Multiplication

Matrix-Vector Multiplication

What if right matrix is a single column?

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We can think of $[A\mathbf{b}_1 \ A\mathbf{b}_2 \ \dots \ A\mathbf{b}_p]$ as collection of simultaneous matrix-vector multiplications

Matrix "Interface"

multiplication

what does AB mean when A and B are matrices?

addition

what does $A + B$ mean when A and B are matrices?

scaling

what does cA mean when A is matrix and c is a real number?

Matrix "Interface"

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what does cA mean when A is matrix and c is a real number?

These should be consistent with matrix-vector interface and vector interface

Matrix Addition

$$[\mathbf{a}_1 \quad \dots \quad \mathbf{a}_n] + [\mathbf{b}_1 \dots \quad \mathbf{b}_n] = [(\mathbf{a}_1 + \mathbf{b}_1) \quad \dots \quad (\mathbf{a}_n + \mathbf{b}_n)]$$

Addition is done column-wise (or element-wise)

$$\text{e.g. } \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} (1+2) & (2+3) \\ (3-2) & (4-3) \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & 1 \end{bmatrix}$$

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This is exactly the same as vector addition, but for matrices

Matrix Addition and Scaling

$$c [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] = [c\mathbf{a}_1 \quad c\mathbf{a}_2 \quad \dots \quad c\mathbf{a}_n]$$

Scaling and adding happen element-wise (or, equivalently, column-wise)

$$\text{e.g. } 2 \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 2(1) & 2(2) \\ 2(-1) & 2(3) \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ -2 & 6 \end{bmatrix}$$

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This is exactly the same as vector scaling, but for matrices

Algebraic Properties (Addition and Scaling)

$$A + B = B + A$$

$$(A + B) + C = A + (B + C)$$

$$A + 0 = A$$

$$r(A + B) = rA + rB$$

$$(r + s)A = rA + sA$$

$$r(sA) = (rs)A$$

In these properties A , B , and C are matrices of the same size and r and s are scalars ($\mathbb{R}$)

We need to know/memorize these

Algebraic Properties (Addition and Scaling)

$$A(BC) = (AB)C$$

$$A(B + C) = AB + AC$$

$$(B + C)A = BA + CA$$

$$r(AB) = (rA)B = A(rB)$$

$$I_m A = A = A I_n$$

In these properties A , B , and C are matrices of the appropriate size so that everything is defined, and r is a scalar

We need to know/memorize these

Matrix Multiplication is not Commutative

Important. AB may not be the same as BA

(it may not even be defined)

Question (Conceptual)

Find a pair of 2D linear transformations T_1 and T_2 such that T_1 followed by T_2 is not the same as T_2 followed by T_1

(also find a pair where they are the same)

Answer: Rotation and Reflection

Computational Aspects of Matrix Multiplication

Matrix Operations in Numpy

Let `a` and `b` be 2D numpy arrays and let `c` be a floating point number

» `a @ b` (matrix multiplication)

» `a + b` (matrix addition)

» `c * a` (matrix scaling)

We've seen these, we've used them a bit, we'll use them much more

Analyzing Linear Algebra Algorithms

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We will not use $O(\cdot)$ notation!

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For numerics, we care about number of **F**loating-oint
Operations (FLOPs):

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$2n$ vs. n is very different
when $n \sim 10^{20}$

Dominant Terms

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A function $f(n)$ is ***asymptotically equivalent*** to $g(n)$ if

$$\lim_{i \rightarrow \infty} \frac{f(i)}{g(i)} = 1$$

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A function $f(n)$ is ***asymptotically equivalent*** to $g(n)$ if

$$\lim_{i \rightarrow \infty} \frac{f(i)}{g(i)} = 1$$

for polynomials, they are equivalent to their dominant term

Dominant Terms

the dominant term of a polynomial is the monomial with the highest degree

$$\lim_{i \rightarrow \infty} \frac{3x^3 + 100000x^2}{3x^3} = 1$$

$3x^3$ dominates the function even though the coefficient for x^2 is so large

A Note on Complexity

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Suppose A and B are $n \times n$ matrices

This operations takes n multiplications and n divisions ($2n$ FLOPS total)

Repeating for each entry gives $\sim 2n^3$ FLOPS

A Note on Parallelization

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

The main part of this procedure is highly parallelizable

A Note on Parallelization

```
a = np.array(...)  
b = np.array(...)  
prod = np.zeros([a.shape[0], b.shape[1]])  
for i in range(a.shape[0]):  
    for j in range(b.shape[1]):  
        prod[i, j] = np.dot(a[i], b[:,j])
```

The main part of this procedure is highly parallelizable

One processor per entry gets you to $\sim 2n$ FLOPS

A Note on Libraries

There are lots of considerations when doing linear algebra with computers

Best leave it to experts (or do research in the area)

LAPACK is the state of the art library for matrix operations

numpy uses LAPACK

Summary

We can reason about matrix equations by reasoning directly about properties of linear transformations

Matrix multiplication coincides with composition of linear transformations

There is an algebra of matrices which is consistent with the algebra of vectors